

Predictive Modeling of Customer Booking Behavior

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The `.info()` method gives us a data description, telling us the names of the columns, their data types and how many null values we have. Fortunately, we have no null values. It looks like some of these columns should be converted into different data types, e.g. `flight_day`.

To provide more context, below is a more detailed data description, explaining exactly what each column means:

- `num_passengers` = number of passengers travelling
- `sales_channel` = sales channel booking was made on
- `trip_type` = trip Type (Round Trip, One Way, Circle Trip)
- `purchase_lead` = number of days between travel date and booking date
- `length_of_stay` = number of days spent at destination
- `flight_hour` = hour of flight departure
- `flight_day` = day of week of flight departure
- `route` = origin -> destination flight route
- `booking_origin` = country from where booking was made
- `wants_extra_baggage` = if the customer wanted extra baggage in the booking
- `wants_preferred_seat` = if the customer wanted a preferred seat in the booking
- `wants_in_flight_meals` = if the customer wanted in-flight meals in the booking
- `flight_duration` = total duration of flight (in hours)
- `booking_complete` = flag indicating if the customer completed the booking

Before we compute any statistics on the data, let's do any necessary data conversion

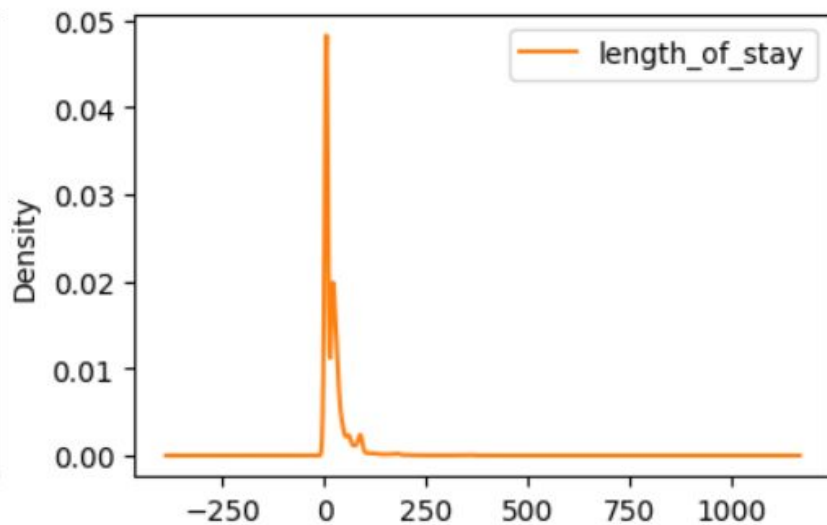
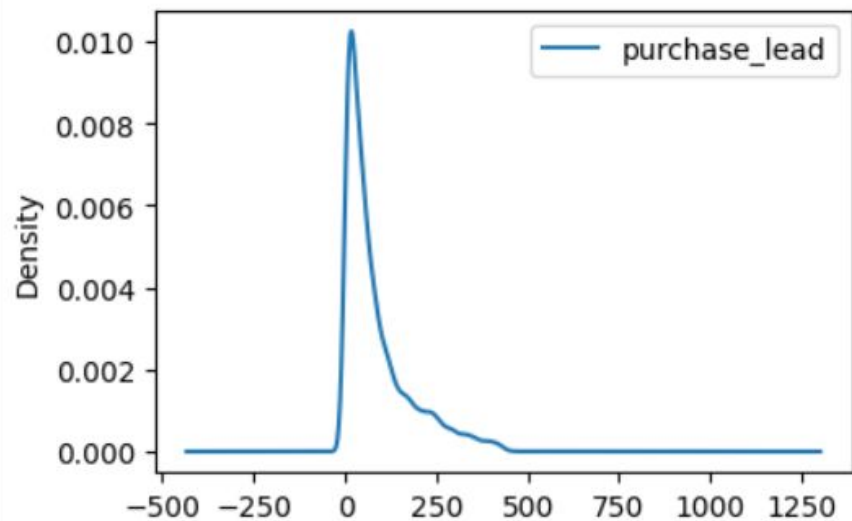
Data Summary:

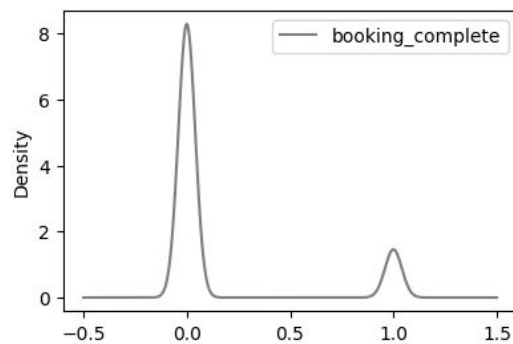
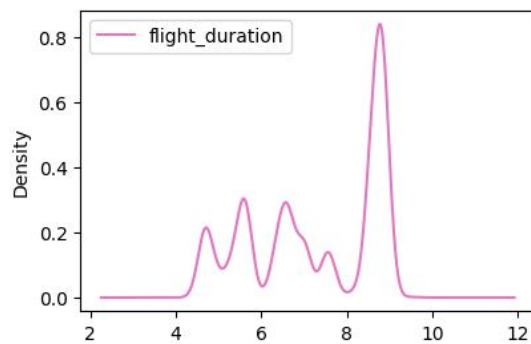
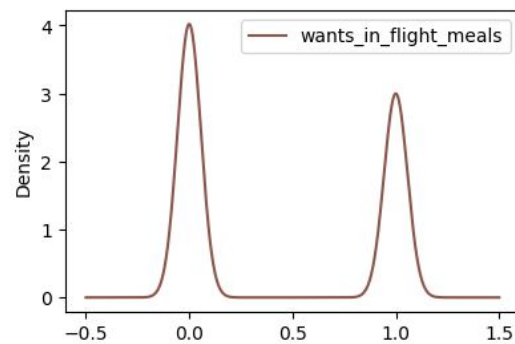
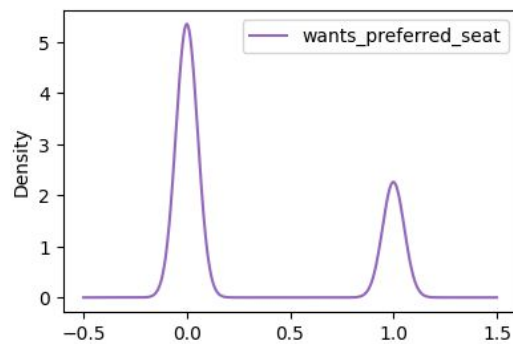
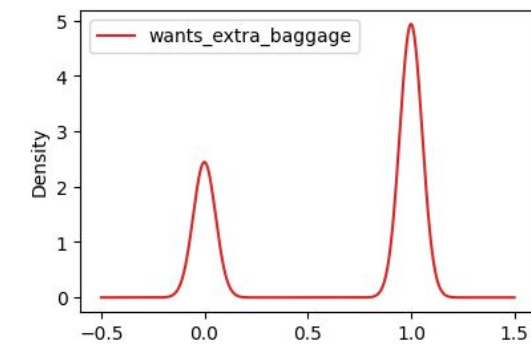
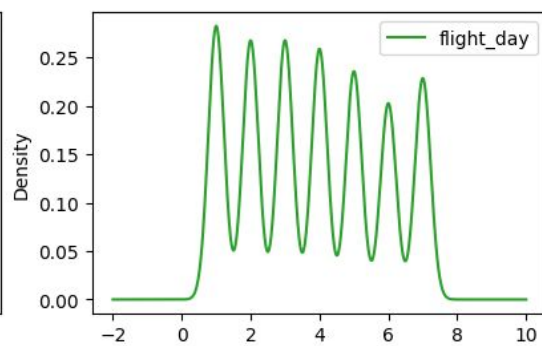
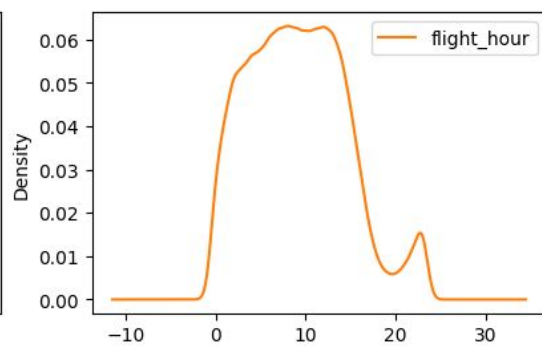
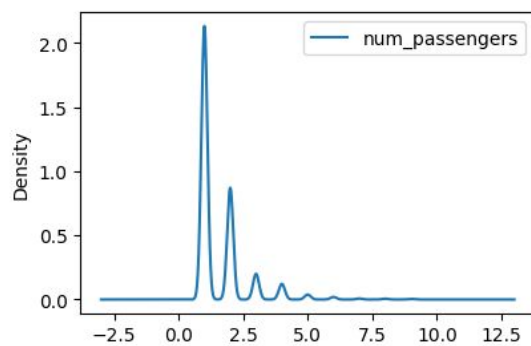
- Dataset: ~50,000 rows of customer booking data
- Target Variable: booking_complete (0 = Not Booked, 1 = Booked)
- Feature Types:
 - Numerical: purchase_lead, flight_hour, flight_duration, etc.
 - Categorical: trip_type, sales_channel, route, booking_origin, etc.

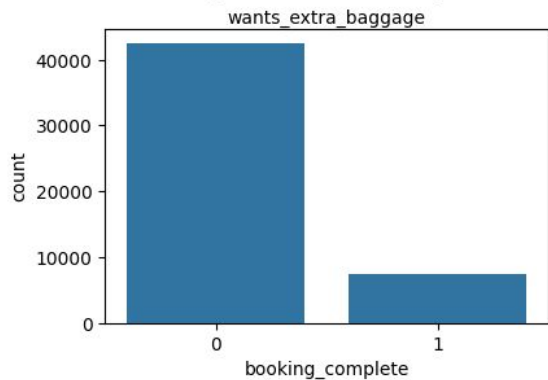
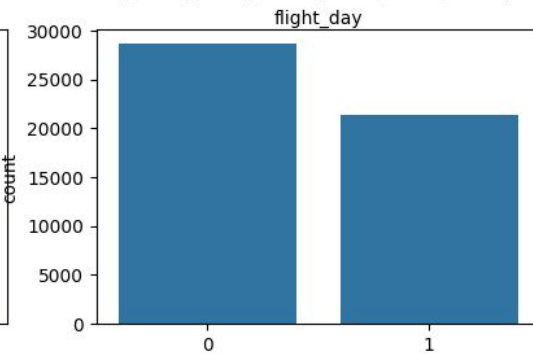
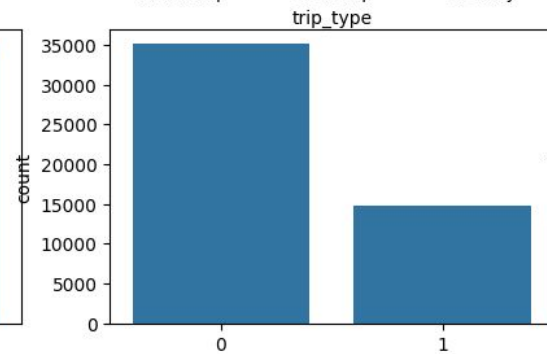
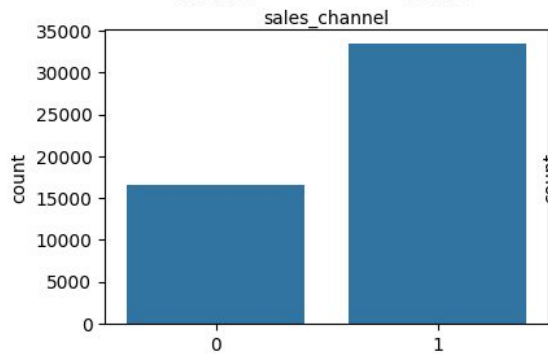
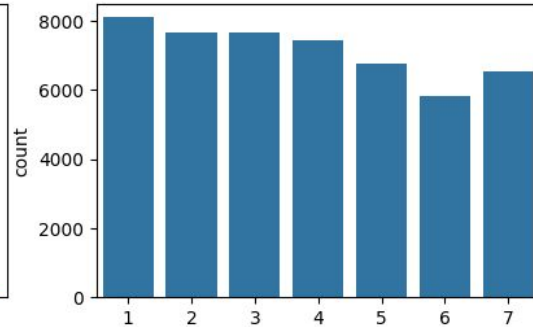
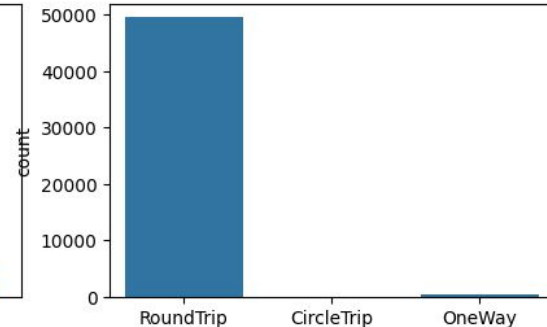
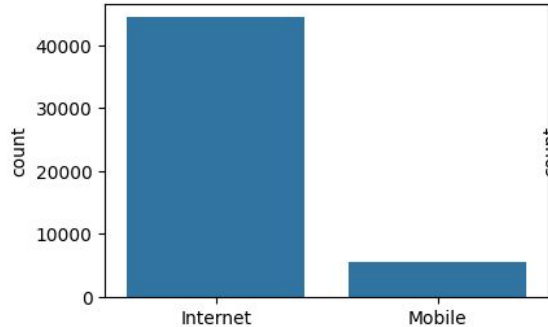


Feature Engineering & Preprocessing:

- Target encoding: route, booking_origin
- One-Hot Encoding: sales_channel, trip_type
- Mapped weekdays to numeric: flight_day
- SMOTE applied to balance target classes
- StandardScaler applied to numerical features









Models Used:

- Logistic Regression (with hyperparameter tuning)
- Random Forest Classifier (with class imbalance handling)



	0.25	1	0.025	0.044	0.04	-0.012	-0.016	0.017	0.0088	-0.00031	0.079	-0.034	0.016	-0.016	0.0091	0.01	-0.013
	-0.13	0.025	1	-0.048	0.0037	-0.17	-0.21	0.28	0.025	0.13	0.23	-0.086	0.082	-0.082	0.004	-0.019	0.015
	0.025	0.044	-0.048	1	0.019	0.0023	0.031	-0.01	0.0099	0.012	-0.024	0.013	-0.023	0.023	-0.00012	0.0045	-0.0039
	0.018	0.04	0.0037	0.019	1	0.0028	0.0022	-0.0021	0.0048	0.0094	0.018	-0.0066	-0.041	0.041	0.0042	-0.0026	0.00021
	0.054	-0.012	-0.17	0.0023	0.0028	1	0.51	-0.017	-0.027	-0.08	-0.22	0.33	-0.032	0.032	-0.04	-0.048	0.061
	0.084	-0.016	-0.21	0.031	0.0022	0.51	1	-0.0019	-0.041	-0.1	-0.32	0.29	-0.066	0.066	-0.029	-0.02	0.031
	0.13	0.017	0.28	-0.01	-0.0021	-0.017	-0.0019	1	0.21	0.22	0.062	0.068	0.054	-0.054	0.012	0.0078	-0.013
	0.048	0.0088	0.025	0.0099	0.0048	-0.027	-0.041	0.21	1	0.32	0.094	0.05	-0.033	0.033	-0.00041	-0.008	0.0072
	0.021	-0.00031	0.13	0.012	0.0094	-0.08	-0.1	0.22	0.32	1	0.15	0.027	0.022	-0.022	-0.003	-0.014	0.014
	-0.059	0.079	0.23	-0.024	0.018	-0.22	-0.32	0.062	0.094	0.15	1	-0.098	0.043	-0.043	0.015	0.038	-0.041
	0.028	-0.034	-0.086	0.013	-0.0066	0.33	0.29	0.068	0.05	0.027	-0.098	1	0.041	-0.041	-0.014	-0.024	0.028
	0.0033	0.016	0.082	-0.023	-0.041	-0.032	-0.066	0.054	-0.033	0.022	0.043	0.041	1	-1	0.012	0.018	-0.022
	-0.0033	-0.016	-0.082	0.023	0.041	0.032	0.066	-0.054	0.033	-0.022	-0.043	-0.041	-1	1	-0.012	-0.018	0.022
	0.0019	0.0091	0.004	-0.00012	0.0042	-0.04	-0.029	0.012	-0.00041	-0.003	0.015	-0.014	0.012	-0.012	1	-0.0043	-0.48
	-0.0047	0.01	-0.019	0.0045	-0.0026	-0.048	-0.02	0.0078	-0.008	-0.014	0.038	-0.024	0.018	-0.018	-0.0043	1	-0.88
	0.0032	-0.013	0.015	-0.0039	0.00021	0.061	0.031	-0.013	0.0072	0.014	-0.041	0.028	-0.022	0.022	-0.48	-0.88	1
num_passengers	purchase_lead	length_of_stay	flight_hour	flight_day	route	booking_origin	extra_baggage	preferred_seat	in_flight_meals	flight_duration	kinos_complete	channel_internet	channel_mobile	type_circletrip	type_oneway	type_roundtrip	

OLS Regression Results

```

=====
Dep. Variable:    booking_complete    R-squared:            0.154
Model:            OLS                 Adj. R-squared:       0.154
Method:           Least Squares       F-statistic:         520.8
Date:             Tue, 22 Jul 2025    Prob (F-statistic):   0.00
Time:             10:37:50            Log-Likelihood:      -12220.
No. Observations: 40000              AIC:                 2.447e+04
Df Residuals:     39985              BIC:                 2.460e+04
Df Model:          14
Covariance Type:  nonrobust
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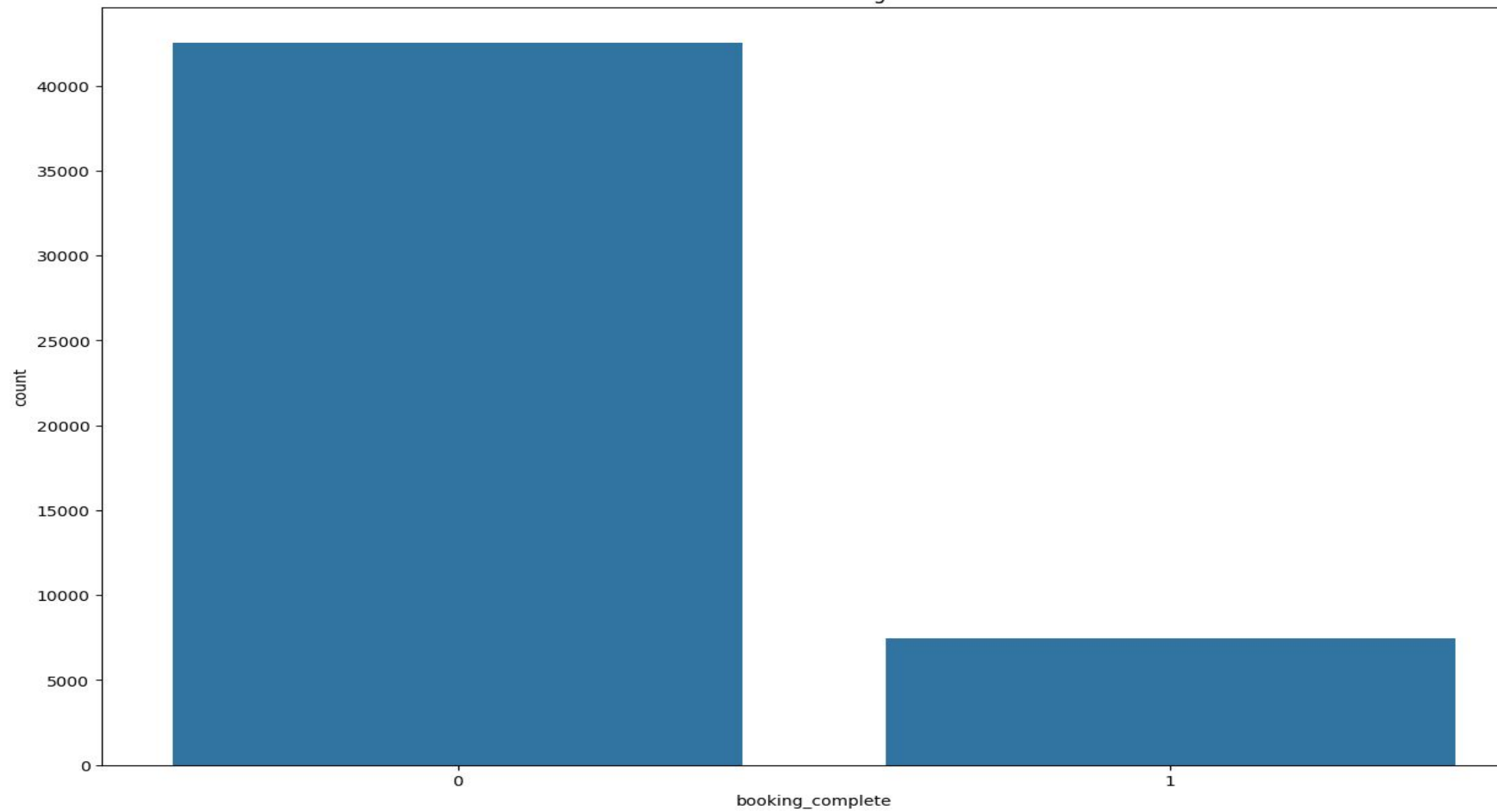
	coef	std err	t	P> t	[0.025	0.975]
-----	-----	-----	-----	-----	-----	-----
const	0.1501	0.002	91.357	0.000	0.147	0.153
x1	-0.0030	0.002	-1.733	0.083	-0.006	0.000
x2	-0.0123	0.002	-7.260	0.000	-0.016	-0.009
x3	-0.0112	0.002	-6.547	0.000	-0.015	-0.008
x4	0.0039	0.002	2.398	0.016	0.001	0.007
x5	-0.0015	0.002	-0.903	0.367	-0.005	0.002
x6	0.0921	0.002	46.568	0.000	0.088	0.096
x7	0.0636	0.002	32.458	0.000	0.060	0.067
x8	0.0186	0.002	10.630	0.000	0.015	0.022
x9	0.0186	0.002	10.603	0.000	0.015	0.022
x10	0.0115	0.002	6.488	0.000	0.008	0.015
x11	0.0047	0.002	2.659	0.008	0.001	0.008
x12	0.0101	0.001	12.163	0.000	0.008	0.012
x13	-0.0101	0.001	-12.163	0.000	-0.012	-0.008
x14	0.0004	0.001	0.256	0.798	-0.003	0.003
x15	-0.0024	0.001	-2.176	0.030	-0.004	-0.000
x16	0.0019	0.001	2.253	0.024	0.000	0.003

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Omnibus:            9554.731    Durbin-Watson:       1.985
Prob(Omnibus):      0.000      Jarque-Bera (JB):    18373.029
Skew:               1.498      Prob(JB):            0.00
Kurtosis:           4.433      Cond. No.            1.61e+16
=====

```

Class Distribution of Target Variable



```

▶ # Apply SMOTE for oversampling the minority class
smote = SMOTE(random_state=42)
X_resampled, y_resampled = smote.fit_resample(X_train, y_train)

# Train a model on the resampled data
model = RandomForestClassifier(random_state=42)
model.fit(X_resampled, y_resampled)

# Predictions and evaluation
y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred))

```

```

↕

```

	precision	recall	f1-score	support
0	0.89	0.91	0.90	12784
1	0.40	0.35	0.37	2216
accuracy			0.83	15000
macro avg	0.64	0.63	0.63	15000
weighted avg	0.82	0.83	0.82	15000

```

[169]
# Define Logistic Regression model with class_weight='balanced'
model = LogisticRegression(class_weight='balanced', random_state=42)

# Train the model
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Evaluate the model
print(classification_report(y_test, y_pred))

```

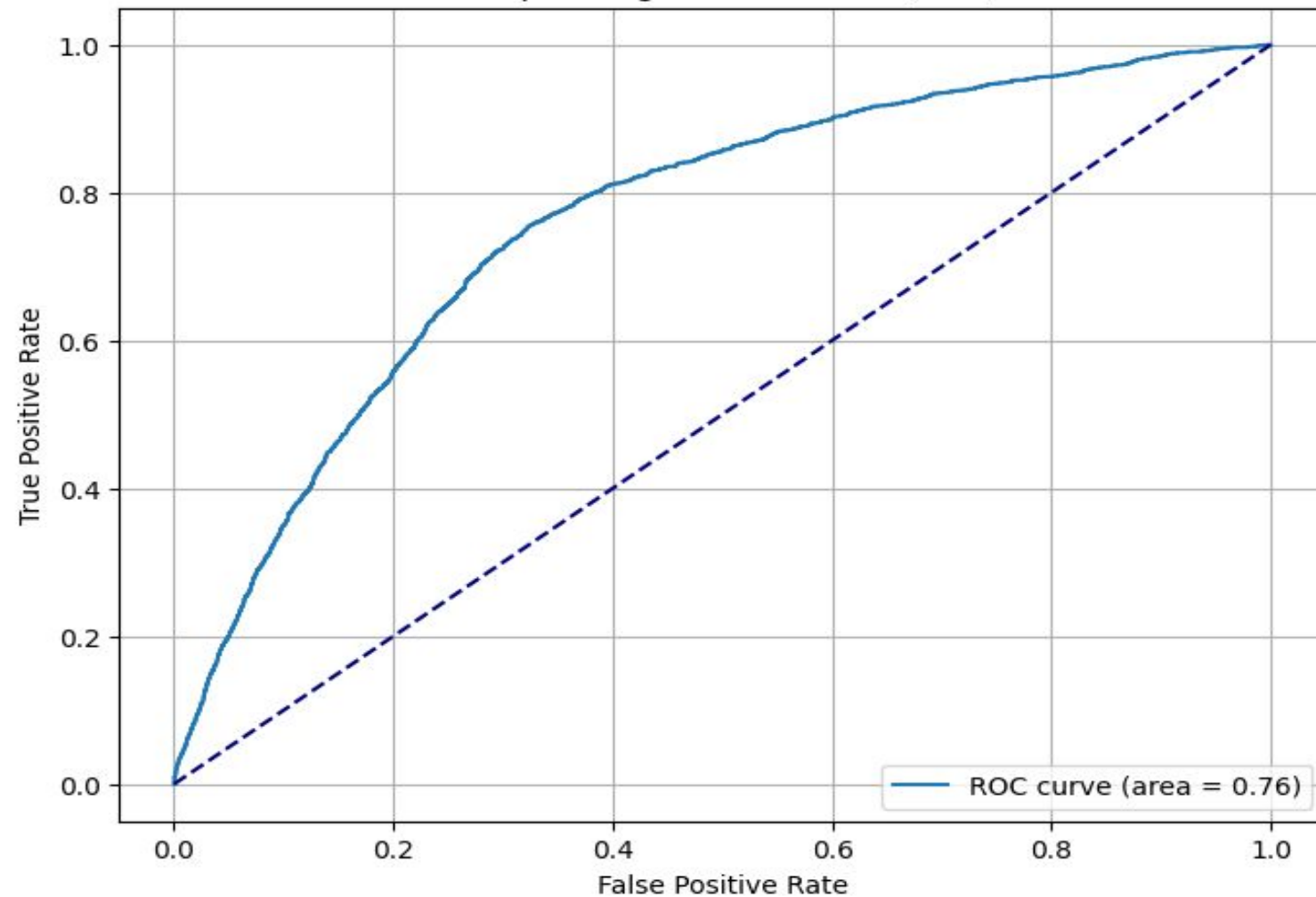
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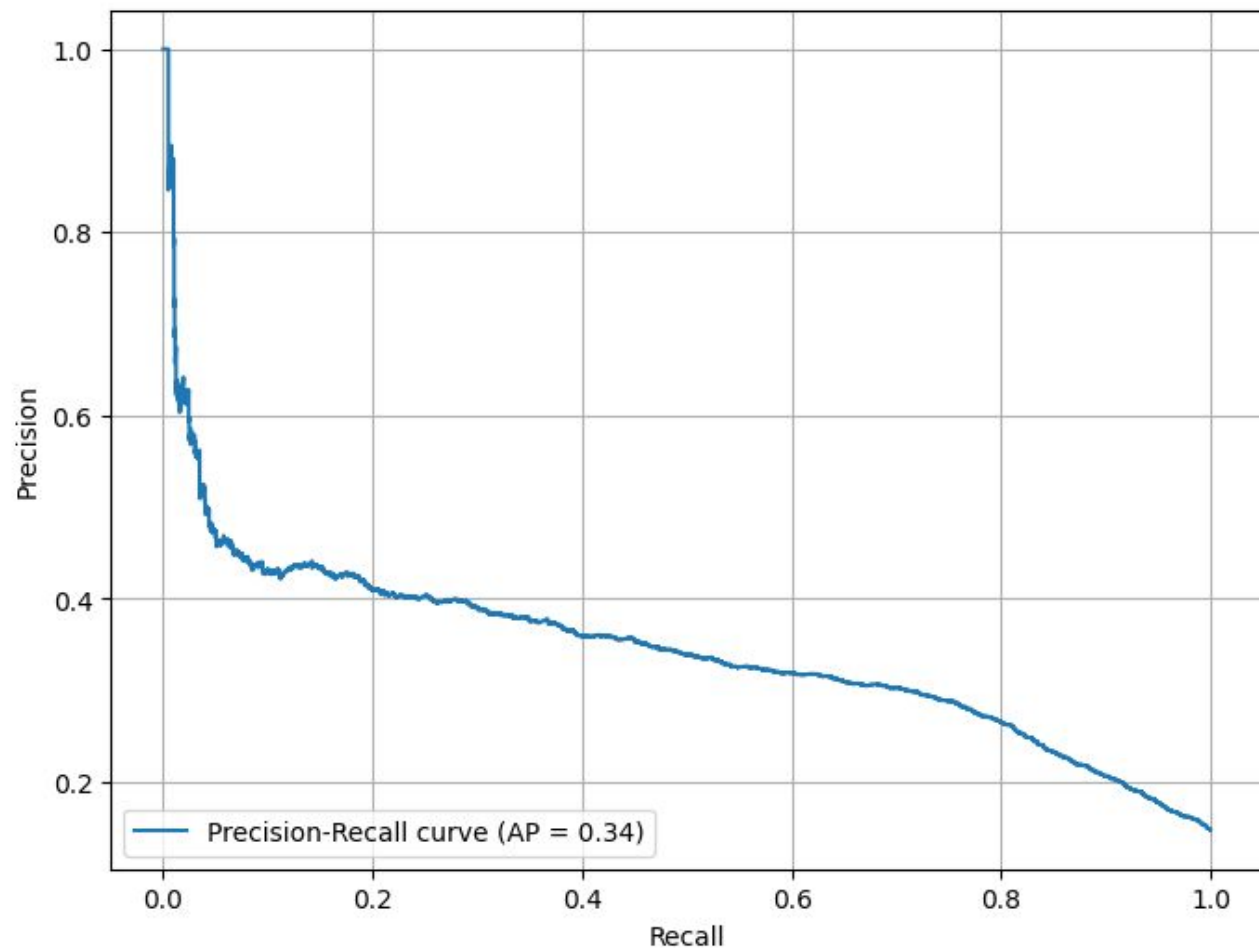
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	precision	recall	f1-score	support
0	0.94	0.70	0.80	12784
1	0.29	0.73	0.42	2216
accuracy			0.70	15000
macro avg	0.62	0.71	0.61	15000
weighted avg	0.84	0.70	0.74	15000

Receiver Operating Characteristic (ROC) Curve

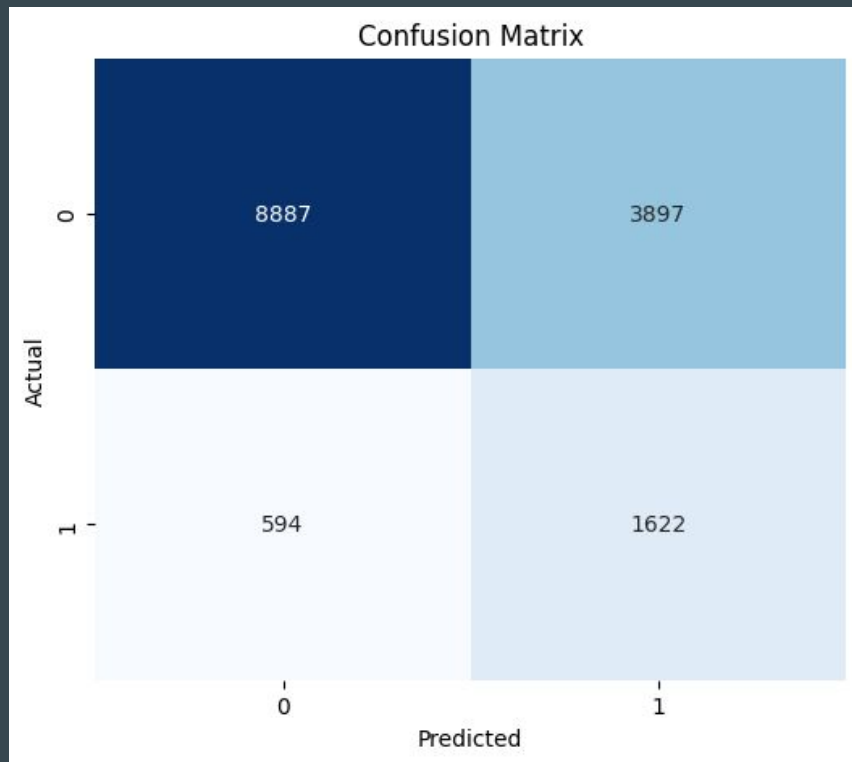


Precision-Recall Curve

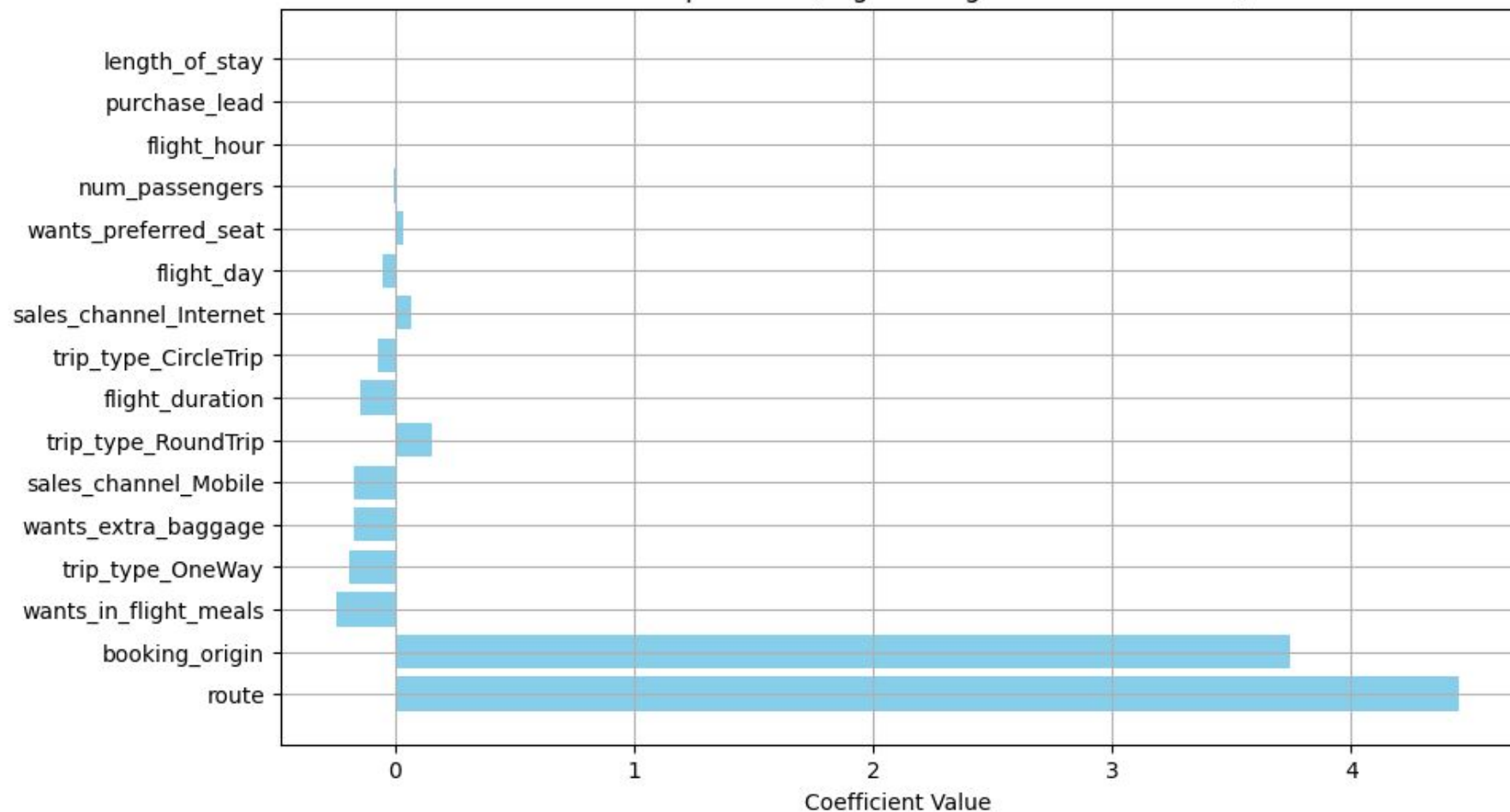


Model Evaluation (Best Performing: Logistic Regression):

- Accuracy: ~83%
- AUC Score: ~0.89
- F1-Score: 0.81



Feature Importance (Logistic Regression Coefficients)



Top Predictive Features:

- Booking origin, Extra baggage, Trip type, Seat selection, Sales channel



Key Insights:

- Extra services indicate higher booking likelihood
- Country of origin and lead time matter
- Strong time feature influence



Important links:

- https://scikit-learn.org/stable/supervised_learning.html
- https://scikit-learn.org/stable/modules/model_evaluation.html
- https://colab.research.google.com/drive/1Uw81UqMsf3Tj8bMHf4K_yvqDbcfR499L#scrollTo=O6XquAe4Byzl

THANK YOU

Any questions?